Enhancing Hospital Efficiency: A SQL Analysis of Hospital Readmission Risks

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I. Purpose

Repeat hospital visits increase healthcare costs, reduce operational efficiency, and may reflect opportunities to improve patient care. This project aims to identify the key factors associated with readmissions and provide actionable recommendations to help reduce preventable readmissions.

II. Executive Summary

This analysis examined 8,000 hospital patient records from 2021–2023 to identify factors associated with 30-day readmissions. The results indicate that advanced age, multiple comorbidities, and previous readmissions are the strongest predictors of future readmission. A high-risk patient cohort defined by these characteristics exhibited a 96.61% readmission rate, providing clear opportunities for targeted intervention and improved discharge planning.

III. Data Overview

Time Period: 2021–2023

Records: 8,000 unique patient records

The original dataset consists of **15 columns**, including patient demographics (age, gender, region), clinical information (primary diagnosis, comorbidities, previous readmissions), treatment details (treatment type, medications prescribed, length of stay), and discharge information. The target variable, **label**, indicates whether a patient was readmitted within **30 days of discharge** (1 = Readmitted, 0 = Not Readmitted).

This analysis focuses on **9 key variables** that are most relevant to evaluating factors associated with 30-day hospital readmissions. Columns that were not directly related to the project objectives were excluded to keep the analysis focused and concise

Key Variables	Description
patient_id	Individual identifier for each patient
age	Patients age range from 18-95
gender	Categorical (male/female)
primary diagnosis	Categorical Diagnosis at time of admissions
comorbidities_count	Numerical Number of additional conditions
length_of_stay	Numerical Number of days in hospital
treatment_type	Categorical Medical / Surgical / Interventional / Conservative
prev_readmissions	Numerical Previous hospital readmissions
Label	Binary 0= No readmissions 1= Readmitted within 30 days

IV. Research Questions

Exploratory Analysis:

1. How serious is the hospital readmission problem?

Business Analysis:

1. Which primary diagnosis are driving readmission?
2. Which patient characteristics are associated with readmissions?
   - Does age influence readmission risk?
   - How do comorbidities affect the likelihood of readmission?
   - Is gender associated with readmission risk?
   - How does a patient's history of previous readmissions influence future readmissions?
3. Which treatments are associated with lower readmission rates?
4. What does a high-risk patient portfolio look like? ?
5. Among high-risk patients, which primary diagnoses and treatment types are associated with the greatest likelihood of readmission?

V. Exploratory Analysis Findings

1. **How serious is the hospital readmission problem?**

Cohort Demographics

- Total Cohort Size: 8,000 unique patients.
- Average Patient age: 57 years old.
- Youngest patient: 18 years old
- Oldest patient: 95 years old
- Average Length of Stay: 8 days.

The Scale of the Readmission Problem

- Total Readmissions: 6,183 patients were readmitted at least once within 30 days.
- Readmission Rate: 77.29%, indicating an exceptionally severe operational and clinical challenge.
- Impact of Stay Duration: Patients who were readmitted averaged a slightly longer initial stay (8 days) compared to those who were not readmitted (7 days).

History of Repeat Admissions

- 94% of the analyzed patients have a history of prior readmissions, meaning only 6% are experiencing their first-ever return.
- The bulk of the issue is concentrated among patients with exactly one or two previous readmissions, representing nearly 82% of the entire cohort.
- Extreme Cases: The highest number of previous readmissions recorded for a single patient was 5.

VI. Business Analysis Findings:

1. Which diagnosis are driving readmission?

Readmission rate by primary diagnosis

- Diabetes and Hypertension drive the highest absolute number of readmissions (nearly 2,000 cases combined) due to sheer patient volume, despite having the lowest rates (~70%).
- High-Risk Diagnoses: Sepsis, COPD, Heart Failure, and Stroke present the highest clinical danger, with readmission rates soaring between 85% and 87%.
- Systemic Issue: Every single diagnosis—even the lowest (Influenza at 68.67%)—exceeds a 68% readmission rate, proving this is a hospital-wide challenge rather than a department-specific one.

primary_diagnosis	Total Patients	Total Readmissions	Readmission Rate (%)
Diabetes	1470	1037	70.54
Hypertension	1301	906	69.64
COPD	952	827	86.87
Stroke	883	753	85.28
Kidney Disease	868	723	83.29
Sepsis	497	432	86.92
Appendicitis	534	380	71.16
Heart Failure	403	346	85.86
Pneumonia	425	314	73.88
Fracture	434	305	70.28
Influenza	233	160	68.67

2. Which patient characteristics are associated with readmissions?

Readmission rate by age group: *Domain Knowledge used to determine age ranges*

age_group	Total Patients	Total Readmissions	Readmission Rate (%)
Older (65+)	2712	2573	94.87
Middle Aged (40-64)	4129	3052	73.92
Young (18-39)	1159	558	48.14

- Readmission risks increase with age.
- The readmission rate nearly **doubles** from the youngest group to the oldest group, suggesting age is strongly associated with hospital readmission in this dataset.
- Middle aged adults make up a majority of the patient population (51.6%).
- Older adults account for a disproportionately high number of readmissions.

Readmission rate by number of comorbidities

comorbidities_count	Total Patients	Total Readmissions	Readmission Rate (%)
10	2	2	100.00
9	13	12	92.31
8	70	68	97.14
7	371	357	96.23
6	1021	982	96.18
5	1960	1799	91.79
4	2397	1799	75.05
3	1520	932	61.32
2	534	208	38.95
1	112	24	21.43

- Positive relationship: For every additional comorbidity, the likelikeness of being readmitted increases
- Patients with 1 to 4 comorbidities have readmission rates ranging from **21.43% to 75.05%**.
- The moment a patient hits **5 comorbidities**, the readmission rate skyrockets to **91.79%** and stays above 90% all the way up to 100%.
- Any patient with 5 or more comorbidities is almost guaranteed to be readmitted. This is the primary target group for aggressive intervention.

Readmission rate by gender

- Men and women are readmitted at almost the exact same rate, with only a tiny 1.27% difference between them (76.62% for men and 77.89% for women). The patient group is split relatively evenly, with women making up 52% of the total and men making up 48%.
- Data suggest that that gender is not a risk factor related to readmission risks.

Readmission rate by number of previous readmissions

- Readmission risk nearly triples from 25.79% (0 previous readmissions) to 69.53% after just a single previous visit.
- Over 81% of all patients (6,557 people) have exactly 1 or 2 previous readmissions, making this group the primary driver of hospital volume.
- Having 2 or more previous readmissions makes a return nearly guaranteed, with rates instantly crossing 90% and hitting 100% at 5 visits.

	prev_readmissions	Total Patients	Total Readmissions	Readmission F (%)
▶	0	477	123	25.79
	1	3800	2642	69.53
	2	2757	2493	90.42
	3	622	593	95.34
	4	320	308	96.25
	5	24	24	100.00

A. **Question 3: Which treatments are associated with lower readmission rates?**

Readmission rate by treatment type

- Interventional treatments are associated with the lowest readmission rate, though only by a very small margin at 76.67%.
- Conservative treatment has the highest readmission rate at 81.73%, making patients who receive this care option the most likely to return to the hospital.
  There is not a notable relationship between treatment type and readmission risk.

B. **Question 4: What does a high-risk patient portfolio look like?**

By combining our demographic, clinical, and behavioral risk thresholds, we built a targeted High-Risk Patient Portfolio defined as:

- Age: 65 or older (age group with highest readmission rate)
- Clinical Complexity: 5 or more comorbidities (threshold where the readmissions rate spikes past 91.7%)
- History: 2 or more previous readmissions (risk spikes to 90.42% at 2+ previous visits)

High Risk Cohort Readmission Rate

- High risk patients account for 2,095 total patients in our dataset. They have a staggering 96.61% readmission rate (resulting in 2,024 readmissions). This successfully isolates the hospital's most critical, near-guaranteed return patients for targeted intervention.

	High-Risk Patients	Readmission Rate (%)
▶	2095	96.61

To gain deeper insights, we analyzed the number of high-risk patients within each primary diagnosis and treatment category.

High risk patients by diagnosis

- Kidney disease (98.50%), sepsis (98.40%), and pneumonia (98.20%) had the highest readmission rates within the high-risk patient cohort, indicating that elderly patients with multiple comorbidities and a history of previous readmissions who present with these diagnoses are at an exceptionally high risk of returning to the hospital.

primary_diagnosis	High-Risk Patients	Readmission Rate (%)
Diabetes	351	94.87
Hypertension	329	96.35
COPD	263	97.34
Stroke	255	97.65
Kidney Disease	200	98.50
Appendicitis	165	93.33
Sepsis	125	98.40
Fracture	123	95.93
Pneumonia	111	98.20
Heart Failure	101	97.03
Influenza	72	97.22

High risk patients by treatment type

- High risk patients receiving conservative treatment have the highest readmission rate.
- Patients receiving medical treatment have the highest number of actual high risk patients.

treatment_type	High-Risk Patients	Readmission Rate (%)
Medical	933	95.82
Surgical	525	98.48
Interventional	432	94.91
Conservative	205	99.02

VII. Recommendations

The findings suggest that hospitals should prioritize resources toward patients with the greatest likelihood of readmission rather than applying the same interventions across all patients. A risk-stratification process should be implemented at admission and discharge to identify patients who are 65 years or older, have five or more comorbidities, or have a history of previous readmissions, as these characteristics were the strongest predictors of future readmission. These patients should receive enhanced discharge planning, comprehensive medication reconciliation, and follow-up appointments scheduled before leaving the hospital. Expanding transition-of-care programs—including home health visits, telehealth check-ins, nurse care coordination, and early post-discharge follow-up—may further reduce preventable readmissions among high-risk patients. Additional attention should be given to patients diagnosed with sepsis, kidney disease, heart failure, pneumonia, and COPD, as these conditions exhibited the highest readmission rates within the high-risk population. Although treatment type showed only modest differences in readmission rates, conservative treatment was associated with the highest readmission rate and should be evaluated to determine whether clinical protocols or follow-up practices can be improved. Finally, hospitals should continuously monitor readmission trends through data-driven dashboards and routinely evaluate intervention outcomes to ensure that resources are directed toward patients who will benefit most, ultimately improving patient outcomes while reducing unnecessary healthcare costs

VIII. Limitations

While this SQL analysis identifies clear risk factors, several limitations in the dataset must be noted. First, the baseline 30-day readmission rate of 77.29% is exceptionally high compared to typical hospital averages of 15% to 25%. This suggests the data is heavily skewed toward a highly vulnerable patient group and may not represent standard, general-admission hospitals. Second, the dataset lacks specific discharge dates, preventing us from analyzing the impact of seasonal spikes or weekend discharge trends. Third, we do not have post-discharge information, such as whether patients actually filled their prescriptions, attended follow-up appointments, or had adequate support at home. Finally, the broad categorization of treatments into general groups (*Interventional, Conservative, Medical*) obscures the specific clinical details and medications that might otherwise explain variations in patient recovery.

IX. Conclusion

In conclusion, this analysis of 8,000 patient records demonstrates that hospital readmissions are highly predictable and concentrated within a specific, vulnerable demographic. While gender and broad treatment types have little impact on return rates, a patient's age, number of comorbidities, and history of past admissions are the strongest indicators of risk. By combining these factors, we identified a high-risk patient cohort—defined as patients aged 65 or older with five or more comorbidities and at least two prior readmissions—that has a staggering 96.61% likelihood of returning to the hospital, especially when diagnosed with kidney disease, sepsis, or pneumonia. To manage this severe 77.29% baseline readmission rate, the hospital must shift from a reactive approach to a targeted, risk-stratified model. By focusing resources, enhanced discharge planning, and transition-of-care programs on this high-risk group, the hospital can improve patient outcomes, optimize bed capacity, and significantly reduce unnecessary operational costs.